Chapter 29 Problem Set

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1a. The sequence a_n converges to $\frac{3}{5}$. Given $\epsilon > 0$, let $N = \frac{1}{25\epsilon} - \frac{7}{5}$. Then $|\frac{3n+4}{5n+7} - \frac{3}{5}| = |\frac{15n-20-15n-21}{25n+35}| = \frac{1}{25n+35} < \frac{1}{25(\frac{1}{25\epsilon}-\frac{7}{5})+35} = \epsilon$.

1b. The sequence converges to 0. We see that $|\frac{n}{n^2+3}| < \epsilon$ when $\epsilon n^2 - n + 3\epsilon > 0$. Since $\epsilon > 0$, this is an upward opening parabola. This means, at some point (which we can call N), all the values of a_n are greater than 0. So $|\frac{n}{n^2} + 3| < \epsilon$ when $n > N$.

1c. The sequence diverges. Given $\epsilon > 0$, we need to prove that there exists N such that if $n > N$ then $\frac{n^2-2}{n+1} > \epsilon$. That is true if $n^2 - \epsilon n - \epsilon - 2 > 0$. Since that is an upward opening parabola, that is true for all n greater than some point, N.

1d. The sequence converges to 0. If $N = \frac{1}{\epsilon}$, then $|\frac{(-1)^n}{n}| = |\frac{1}{n}| < \frac{1}{N} = \epsilon$.

1e. The sequence converges to 0. If $N = \frac{1}{\epsilon}$, then $|\frac{\sin n}{n}| \leq |\frac{1}{n}| < \frac{1}{N} = \epsilon$.

1f. The sequence converges to 0. If $N = \frac{1}{\epsilon}$, then $|\frac{\cos n}{n}| \leq |\frac{1}{n}| < \frac{1}{N} = \epsilon$.

1g. The sequence converges to $\frac{1}{2}$. If $N = \frac{\epsilon}{2} + \frac{1}{2} + \frac{1}{8\epsilon}$, then

$$\left| n - \sqrt{n^2 - n} - \frac{1}{2} \right| < \left| \frac{\epsilon}{2} + \frac{1}{2} + \frac{1}{8\epsilon} - \sqrt{\left(\frac{\epsilon}{2} + \frac{1}{2} + \frac{1}{8\epsilon} \right)^2 - \left(\frac{\epsilon}{2} + \frac{1}{2} + \frac{1}{8\epsilon} \right)} - \frac{1}{2} \right|$$

$$= \left| \frac{\epsilon}{2} + \frac{1}{2} + \frac{1}{8\epsilon} - \sqrt{\frac{\epsilon^2}{4} - \frac{1}{8} + \frac{1}{64\epsilon^2}} - \frac{1}{2} \right| \leq \left| \frac{\epsilon}{2} + \frac{1}{2} + \frac{1}{8\epsilon} + \frac{\epsilon}{2} - \frac{1}{8\epsilon} - \frac{1}{2} \right| = \epsilon.$$

2a. Using limit laws, we have $\lim_{n\to\infty}(1 + \frac{n^2}{n^2+3})(3 + \frac{n}{n^2+6}) = \lim_{n\to\infty}(1 + \frac{n^2}{n^2+3}) \cdot \lim_{n\to\infty}(3 + \frac{n}{n^2+6}) = (1 + \lim_{n\to\infty} \frac{n^2}{n^2+3})(3 + \lim_{n\to\infty} \frac{n}{n^2+6})$.

Now, let's show that $\lim_{n\to\infty} \frac{n^2}{n^2+3} = 1$. We know that $|\frac{n^2}{n^2+3} - 1| = \frac{3}{n^2+3} < \epsilon$ if $\epsilon n^2 + 3n - 3 > 0$. Since it is an upward opening parabola in n, there exists N such that $|\frac{n^2}{n^2+3} - 1| < \epsilon$.

Now, we show that $\lim_{n\to\infty} \frac{n}{n^2+6} = 0$. We can see that $\epsilon n^2 - n + 6\epsilon > 0$ implies $|\frac{n}{n^2+6}| < \epsilon$, and that is true for all n greater than some N because it represents an upward opening parabola.

Therefore, $(1 + \lim_{n\to\infty} \frac{n^2}{n^2+3})(3 + \lim_{n\to\infty} \frac{n}{n^2+6}) = (1+1)(3+0) = 6$.

2b. We see that $\lim_{n\to\infty} \frac{\sin^3 n + \cos n}{n^3+3n^2+1} = \lim_{n\to\infty} \frac{\sin^3 n}{n^3+3n^2+1} + \lim_{n\to\infty} \frac{\cos n}{n^3+3n^2+1}$. Also, $-\frac{1}{n^3+3n^2+1} \leq \frac{\sin^3 n}{n^3+3n^2+1} \leq \frac{1}{n^3+3n^2+1}$.

Now, let's show that $\frac{1}{n^3+3n^2+1}$ converges. We know that $|\frac{\sin^3 n}{n^3+3n^2+1} - 0| \leq |\frac{1}{n^3+3n^2+1} - 0| < \epsilon$ if $\epsilon n^3 + 3\epsilon n^2 + \epsilon - 1 > 0$, which is true for all n greater than some N because it is a cubic with a positive leading coefficient.

Similarly for $\frac{\cos n}{n^3+3n^2+1}$, we get that $\lim_{n\to\infty} \frac{\sin^3 n + \cos n}{n^3+3n^2+1} = 0 + 0 = 0$.

3.

4. Assume for contradiction that $C < s$. This implies that $a_n < C < s$ for all n. If we set $\epsilon = s - C$, then $|a_n - s| = s - a_n < s - C$, which means $a_n > C$, which is a contradiction.

We can't conclude that $s < C$ because all the terms might tend to C but never reach it. For example, consider the sequence $1 - \frac{1}{2^n}$. It converges to 1, but the least upper bound is also 1. So $s = C$.

For contradiction, assume that $C < s$. So $a_n \leq C < s$ for all n. If $\epsilon = s - c$, then $|a_n - s| = s - a_n < s - C$, so $a_n > C$, which is a contradiction.

5. Call the sequence a_n. Since it converges (say to s), for all $\epsilon > 0$ there exists N such that if $n > N$ then $|a_n - s| < \epsilon$. The values of a_n for $n \leq N$ form a finite set, which has a minimum and a maximum, say A and B respectively. For $n > N$, a lower bound is $s - \epsilon$ and an upper bound is $s + \epsilon$. So a lower bound of a_n is $\min(A, s - \epsilon)$ and an upper bound is $\max(B, s + \epsilon)$.

An example of a bounded sequence that is not convergent is $(-1)^n$. It is always between -1 and 1 but does not converge.

6. Let's construct a monotone subsequence, b_n, of a_n assuming it is bounded. Define $b_1 = a_1$. If there are infinitely many elements in a_n that are $\geq b_n$ for all n, let b_{i+1} be the first element after b_i that is $\geq b_i$. If there aren't infinitely many elements in a_n that are $\geq b_n$ for some n, then at one point there are infinitely many elements in a_n that are less than that element. In that case, let b_{i+1} be the first element after b_i that is $\leq b_i$. This results in either a monotonically increasing or decreasing sequence.

7. Assume that a monotonically increasing sequence a_n is bounded, and call the supremum L. By the definition of a supremum, for every $\epsilon > 0$, there exists N such that $L - \epsilon < a_N \leq L$. Since a_n is increasing, for all $n \geq N$, we have $a_N \leq a_n \leq L$, so $L - \epsilon < a_n < L + \epsilon$, or equivalently $|a_n - L| < \epsilon$. The proof is similar for monotonically decreasing sequences.

Now, assume that a monotonically increasing sequence a_n is convergent, and call the limit L. That means there exists N such that for all $n > N$ we have $|a_n - L| < \epsilon$. Since a_n is increasing, we have that $a_k < L + \epsilon$ for all $k \in \mathbb{N}$. Also, the lower bound is a_1 since it's increasing. The proof is similar for monotonically decreasing sequences.